Marking Scheme
Strictly Confidential
(For Internal and Restricted use only)
Secondary School Supplementary Examination, July 2024
SUBJECT NAME: SCIENCE **SUBJECT CODE: 086**
PAPER CODE: 31/S/1

General Instructions: -

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SECONDARY SCHOOL SUPPLEMENTARY EXAMINATION, July 2024

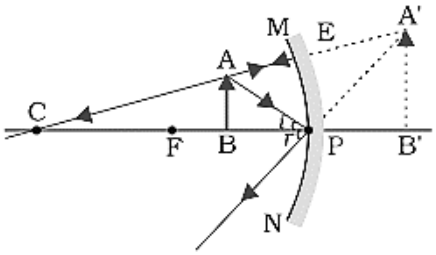
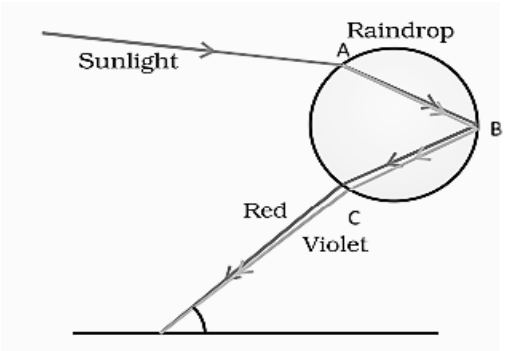
MARKING SCHEME

CLASS : X SCIENCE (Subject Code–086)

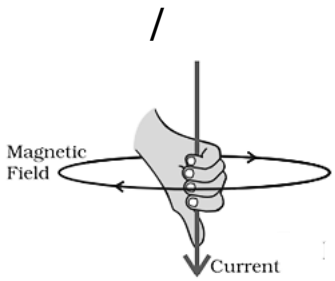
[Paper Code: 31/S/1]

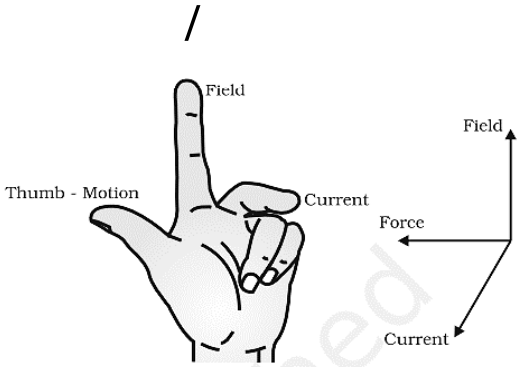
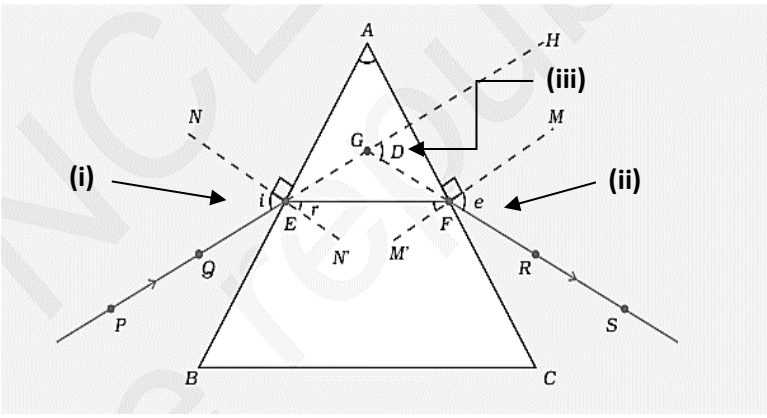
Maximum Marks: 80

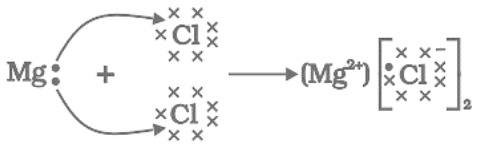
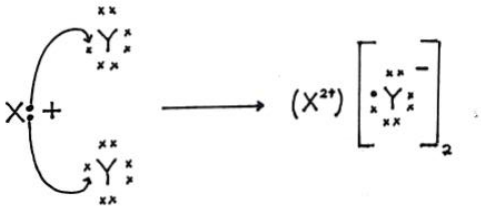
Q. No.	EXPECTED ANSWERS / VALUE POINTS	Marks	Total Marks
	SECTION A		
1.	(A)	1	1
2.	(C)	1	1
3.	(B)	1	1
4.	(C)	1	1
5.	(D)	1	1
6.	(C)	1	1
7.	(B)	1	1
8.	(C)	1	1
9.	(D)	1	1
10.	(C)	1	1
11.	(A)/(C) * Any one is correct as the diagram is not clear	1	1
12.	(B)	1	1
13.	(D)	1	1
14.	(C)	1	1
15.	(B)	1	1
16.	(C)	1	1
17.	(A)	1	1
18.	(B)	1	1
19.	(C)	1	1
20.	(B)	1	1
	SECTION B		
21.	• Ductility: ability of metals to be drawn into thin wires. • Malleability: ability of metals to be beaten into thin sheets. • Gold/silver/aluminium/iron (or any other correct metal) • Metal can be drawn into wires as well as beaten into thin sheets / example of use of any metal exhibiting these properties in daily life.	½ ½ ½ ½	2
22.	• Two ways 1. Errors in DNA copying mechanism	½	

	2. Sexual reproduction • Importance: (any one of following) 1. Helps a species to survive 2. Helps an organism to adapt to the changing environment 3. Forms the basis of evolution (or any other)	½ 1	2
23.	(a) • X – Sensory neuron • Y – Motor neuron • Functions: X: to carry message from skin/sense organs to brain / spinal cord Y: to carry response from brain/spinal cord to muscle / body parts **OR** (b) (i) Cerebellum/Hind brain (ii) Medulla/Hind brain (iii) Fore-brain (iv) Cerebrum/ Fore-brain	½ ½ ½ ½ ½ × 4	2
24.	 [P : Pole, F : Principal Focus, AB : Object, A′B′: Image] *[Diagram from NCERT fig 9.7(f) page 140]* Ray diagram Labelling of pole and principal focus Direction of rays	1 ½ ½	2
25.	(a)  		

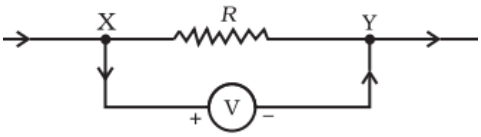
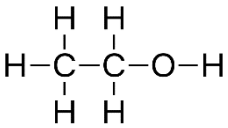
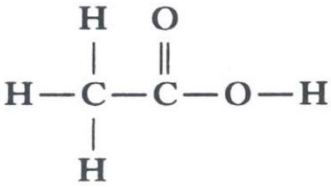
	[Fig 10.8 page 16] Diagram Correct labelling of A,B and C **OR** (b) • Ciliary muscles • Ciliary muscles adjust/change the focal length of eye lens by changing the curvature of eye lens.	 ½ 1½ 1 1	 2
26.	A vegetarian food habit involves only two step food chain Producer → Primary consumer(Herbivore) An average of 10% of the amount of organic matter reaches to the next level of consumer, hence it gets more energy.	2	2
	SECTION C		
27.	(a) $Fe(s) + CuSO_4(aq) \rightarrow FeSO_4(aq) + Cu(s)$ (Copper sulphate) (Iron sulphate) Blue Light Green (any other suitable example) (b) • $CuO(s) + H_2(g) \xrightarrow{Heat} Cu(s) + H_2O(g)$ • Redox reaction	 1 1 1	 3
28.	(a) • pH of a solution is a measure of H^+ ion concentration in it / a number which indicates the acidic or basic nature of a solution. • City B • Lower the pH, more is the acidic nature. • Acidic rain water lowers the pH of pond making the survival of aquatic life difficult. **OR** (b) (i) (I) Acid – HCl / Hydrochloric acid Base – NaOH/Sodium hydroxide (II) Acid – H_2SO_4/ Sulphuric acid Base – NH_4OH/ Ammonium Hydroxide (ii) • Neutral Reason : salt of strong acid and strong base	 1 ½ ½ 1 ½ ½ ½ ½	

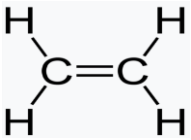
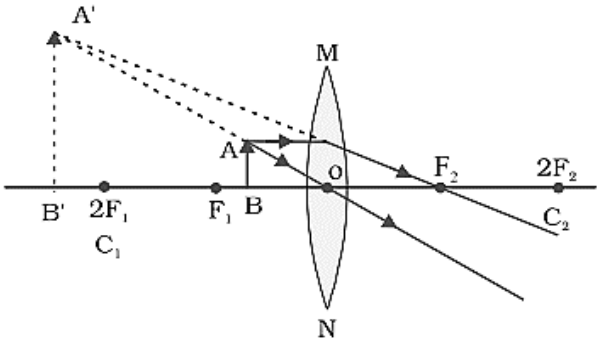
	• Acidic Reason : salt of strong acid and weak base.	½ ½	3
29.	(a) (i) Sepals (ii) Petals (iii) Anther (iv) Ovule. (b) • Through the pollen tube. • Male germ cell fuses with female germ cell to produce a zygote/ fertilization takes place	½ ½ ½ ½ ½ ½	3
30.	(a) • Dominant trait, • Bb (b) (i) 75% (ii) 25% • Two copies of the recessive traits(genes) are required for its expression / reappearance.	 ½ ½ ½ ½ 1	3
31.	(i) R = 2 Ω + 4 Ω + 6 Ω = 12 Ω (ii) $I = \frac{V}{R} = \frac{6V}{12\Omega} = 0.5$ A (iii) V= IR = 0.5A × 4Ω = 2V	1 1 1	3
32.	(a) (i) Imagine that you are holding a current-carrying straight conductor in your right hand such that the thumb points towards the direction of current. Then your fingers will wrap around the conductor in the direction of the field lines of the magnetic field. This is known as the right-hand thumb rule.  *[Fig 12.7, page 200 of NCERT]* ***Note : Award full marks if explained diagrammatically.*** (ii) Stretch the thumb, forefinger and middle finger of your left hand such that they are mutually perpendicular (Fig. 13.13). If the first finger points in the direction of magnetic field and the second finger in	1	

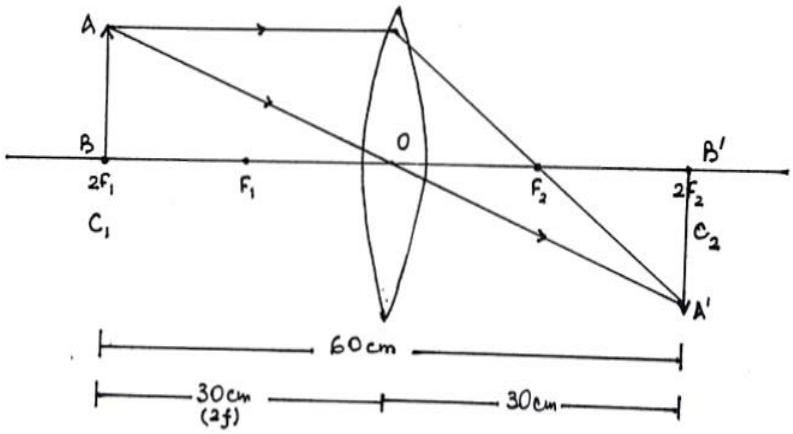
	the direction of current, then the thumb will point in the direction of motion or the force acting on the conductor.  *[Fig 12.13, page 203 of NCERT]* ***Note : Award full marks if explained diagrammatically.*** (b) Out of plane of the paper.	1 1	3
33.	•  Labelled diagram *[Fig 10.4, page 166 of NCERT]* • White light will split into a band of seven colours of spectrum. • Different colours of white light bend / refract through different angles with respect to incident ray as they pass through a prism.	1 ½ ½ 1	3
	SECTION D		
34.	(a) • X : 2,8,2 • Y: 2,8,7 • (Any one of the following below formations)	½ ½	

	 /  • Name: Magnesium chloride • Formula : $MgCl_2$/XY_2 • The compound will dissociate into ions/ions will move freely to the opposite electrodes/electrolysis will occur. • (i) Cathode: Mg^{2+} / Magnesium ion/Cation (ii) Anode: Cl^- /Chloride ion/Anion **OR** (b) (i) • Zinc, Iron, Lead, Copper (any two) • Zinc is more reactive than Iron/Iron is more reactive than Copper/Lead is more reactive than Copper (or any other suitable combination) • Demonstration: 1. Put iron nail in a solution of copper sulphate taken in a test tube. 2. After sometime blue colour of $CuSO_4$ fades. (Any other relevant activity) (ii) • Calcination • Reduction using Carbon Chemical Reactions 1. $ZnCO_3 \xrightarrow{Heat} ZnO + CO_2$ 2. $ZnO + C \longrightarrow Zn + CO$	1 ½ ½ 1 ½ ½ ½ , ½ 1 1 ½ ½ ½ ½	5
35.	(a) (i) A – Mouth/buccal cavity B – Stomach C – Pancreas Enzymes A – Salivary Amylase B – Pepsin	½ ½ ½ ½ ½	

	C – Trypsin/Lipase (ii) • Digested food is taken up by the walls of the small intestine/ villi • The food is absorbed here • Blood vessels in the villi take absorbed food to each and every cell of the body. • Utilisation of this food is done to obtain energy, building up new tissues and repair of old tissues. (if explained in paragraph award full marks) **OR** (b) (i) A – Pulmonary artery B – Aorta C – Capillaries **Functions:** A – Carries deoxygenated blood from heart to lungs. B – Carries oxygenated blood to different parts of the body. C - Exchange of material between blood and surrounding parts. (ii) • To prevent oxygenated and deoxygenated blood from mixing • This allows a highly efficient supply of oxygen to the body of birds to maintain their body temperature.	½ 2 ½ ½ ½ ½ ½ ½ 2	5
36.	(a) (i) • R= $\rho \frac{l}{A}$ $\rho = \frac{R \times A}{l}$ • SI unit of $\rho = \frac{\text{SI unit of R} \times \text{SI unit of A}}{\text{SI unit of l}} = \frac{\Omega \times m^2}{m}$ $= \Omega$ m (ii)R= 3Ω ; l =120 cm = 1.2 m ; d = 0.4 mm or r = 0.2 mm $\rho = \frac{R \times A}{l}$ A= $3.14 \times (0.2mm)^2 = 3.14 \times 4 \times 10^{-8} m^2$ $= 3\Omega \times \frac{3.14 \times 4 \times 10^{-8} \ m^2}{1.2m}$ $= 3.14 \times 10^{-7}$ Ω m	 1 1 ½ ½ 1 1	

	OR (b) (i) • When electric current is passed through a resistor electric energy is dissipated and appears as heat energy. • When charge Q flows through a conductor of resistance R having potential difference of V , the workdone W is given by-  W= V x Q = V x I x t = IR x I x t = I^2Rt (∵V = IxR – Ohm's Law) When circuit is purely resistive electric energy is converted into heat energy. H= I^2Rt (ii) Here I= 2A ; R= 10Ω ; t= 100s H= I^2Rt = $(2A)^2$ x 10Ω x 100s = 4000 J	1 1 1 1 1	5
	SECTION E		
37.	(a)  / CH_3CH_2OH (b) Carbon dioxide and water / CO_2 and H_2O (c) (i) • Ethanol is oxidised to ethanoic acid. •  / CH_3COOH • Acts as an oxidizing agent.	1 ½, ½ ½ 1 ½	

	OR (c) (ii) • Ethene/ C_2H_4 is formed. •  / $CH_2 = CH_2$ • Acts as a Dehydrating agent.	½ 1 ½	4
38.	(a) Sexual – Seed Asexual - Roots/stem/leaves. . (b) (i) Rose/Jasmine (or any other) (ii) Grapes/orange/banana (or any other) (c) (i) (Any two of the following) • Production of genetically similar plants • Propagated plants bear fruits and vegetables earlier than those produced from seeds. • Propagation of plants that have lost capacity to produce seeds. (Any other) **OR** (c) (ii) Put the small pieces of potato containing notches or buds in moist cotton for few days and observe. Notches or buds will give rise to fresh green shoots and roots.	½ ½ ½ ½ 1, 1 2	4
39	(a) More than 2cm and less than 4cm from the optical centre of the lens (b) Less than 6cm from the optical centre of the lens (c) (i)  	1 1 2	

	OB = 12 cm OF = 18 cm A′B′ : Image *[Fig. 9.16 (f) page 155 NCERT]* ***(Deduct ½ mark if direction of rays is not shown)*** **OR** (c) (ii) As per the information given in the question, the lens lies at the centre of the object placed and image formed.  The position of the object is at $2F_1$ therefore object distance = image distance = 2f ∴ 2f + 2f = 60 cm 4f = 60 f = 15cm ∴ The focal length of the lens = 15 cm **Note : Do not deduct marks if diagram is not drawn.**	1 1	4

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MARKING SCHEME

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[Paper Code: 31/S/2]

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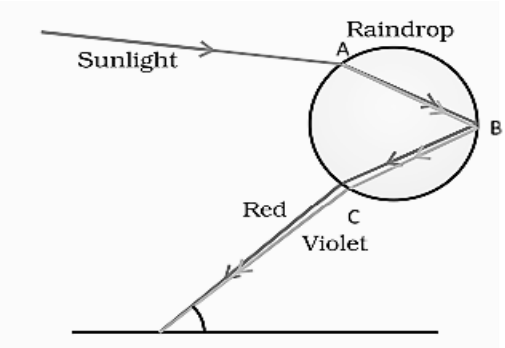
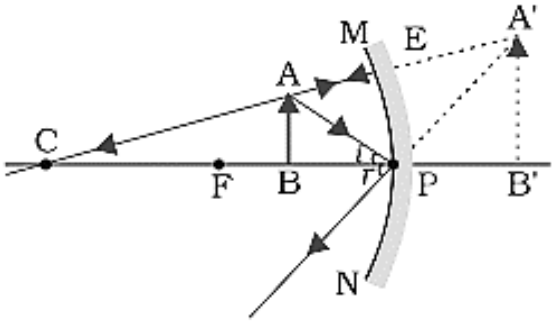
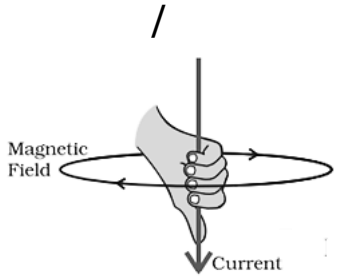
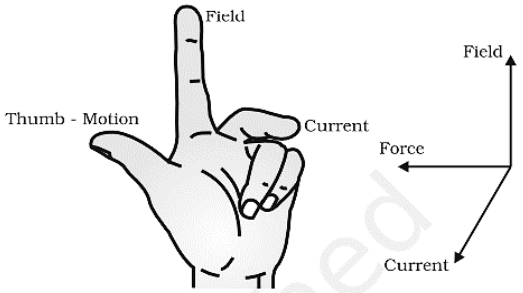
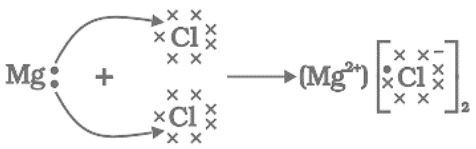
8	(D)	1	1
9	(A)/(C)	1	1
10	(D)	1	1
11	(D)	1	1
12	(D)	1	1
13	(B)	1	1
14	(C)	1	1
15	(C)	1	1
16	(C)	1	1
17	(B)	1	1
18	(C)	1	1
19	(A)	1	1
20	(B)	1	1
	SECTION B		
21.	• Solutions B and C • Due to the presence of free ions.	½, ½ 1	2
22.	(a) • X – Sensory neuron • Y – Motor neuron • Functions: X : to carry message from skin/sense organs to brain / spinal cord Y : to carry response from brain/spinal cord to muscle / body parts	 ½ ½ ½ ½	

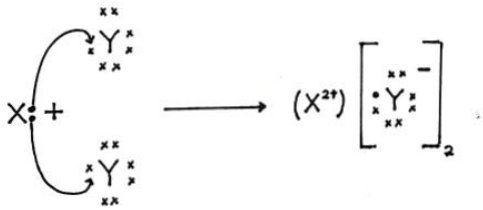
	OR **(b)** (i) Cerebellum/Hind brain (ii) Medulla/Hind brain (iii) Fore-brain (iv) Cerebrum/ Fore-brain	½ × 4	2
23.	• Two ways 1. Errors in DNA copying mechanism 2. Sexual reproduction • Importance: (any one of following) 1. Helps a species to survive 2. Helps an organism to adapt to the changing environment 3. Forms the basis of evolution (or any other)	 ½ ½ 1	2
24.	(a)  *[Fig 10.8 page 16]* Diagram Correct labelling of A,B and C **OR** (b) • Ciliary muscles • Ciliary muscles adjust/change the focal length of eye lens by changing the curvature of eye lens.	 ½ 1½ 1 1	2

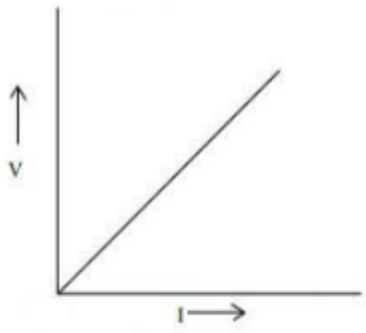
25.	 [P: Pole, F: Principal Focus, AB : Object, A′B′: Image] *[Diagram from NCERT fig 9.7(f) page 140]* Ray diagram Labelling of pole and principal focus Direction of rays	 1 ½ ½	 2
26.	• Phytoplankton ➡ Zooplankton ➡ Small fish ➡ Large fish • Primary Consumer: - Zooplankton • Tertiary Consumer: - Large fish (or Any other Example of aquatic food chain)	1 ½ ½	2
	SECTION C		
27.	(a) • Dominant trait, • Bb (b) (i) 75% (ii) 25% Two copies of the recessive traits (genes) are required for its expression / reappearance.	 ½ ½ ½ ½ 1	3
28.	(a) (i) • CO_2 • Lime water turns milky. • $Ca(OH)_2 + CO_2 \rightarrow CaCO_3 + H_2O$ (ii) • Milky colour turns to colourless in excess of CO_2. • Formula: -$Ca(HCO_3)_2$ **OR** (b) • $NaCl + H_2O + CO_2 + NH_3 \rightarrow NaHCO_3 + NH_4Cl$ Baking soda • Baking soda is mixed with a mild edible acid such as tartaric acid. • When baking powder is heated or mixed in water then Carbon dioxide produced during the reaction makes bread or cake to rise.	 ½ ½ 1 ½ ½ 1 ½ ½	

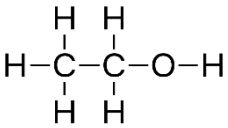
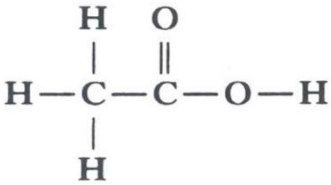
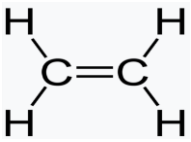
	• $NaHCO_3 + H^+ \longrightarrow CO_2 + H_2O$ + Sodium salt of acid	1	3
29	(a) Function of Ovary: (any one function) • Formation of female germ cells or eggs. • Secretion of Oestrogen. Function of Testes: - (any one function) • Formation of male germ cells/sperms. • Secretion of testosterone. (b) (i) Sperm formation requires a lower temperature than the normal body temperature. (ii) To receive, nurture and nourish the growing embryo.	½ ½ 1 1	3
30.	(a) $Fe(s) + CuSO_4(aq) \rightarrow FeSO_4(aq) + Cu(s)$ (Copper sulphate) (Iron sulphate) Blue Light Green (any other suitable example) (b) • $CuO(s) + H_2(g) \xrightarrow{Heat} Cu(s) + H_2O(g)$ • Redox reaction	1 1 1	3
31.	(a) (i) Imagine that you are holding a current-carrying straight conductor in your right hand such that the thumb points towards the direction of current. Then your fingers will wrap around the conductor in the direction of the field lines of the magnetic field. This is known as the right-hand thumb rule.  *[Fig 12.7, page 200 of NCERT]* ***Note : Award full marks if explained diagrammatically.*** (ii) Stretch the thumb, forefinger and middle finger of your left hand such that they are mutually perpendicular . If the first finger points in the	1	

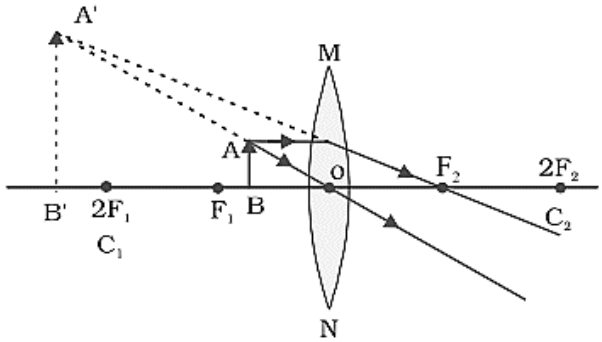
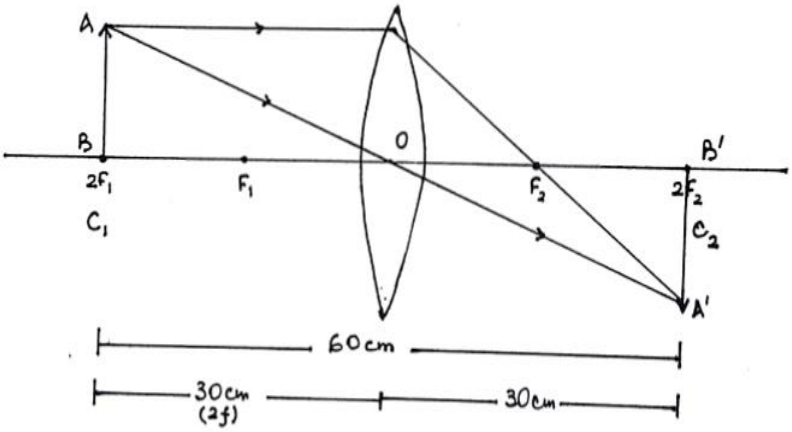
	direction of magnetic field and the second finger in the direction of current, then the thumb will point in the direction of motion or the force acting on the conductor.  *[Fig 12.13, page 203 of NCERT]* ***Note : Award full marks if explained diagrammatically.*** (b) Out of plane of the paper.	1 1	3
32.	(i) R = 2 Ω + 4 Ω + 6 Ω = 12 Ω (ii) $I = \frac{V}{R} = \frac{6V}{12\Omega} = 0.5$ A (iii) V= IR = 0.5A × 4Ω = 2V	1 1 1	3
33	**(a)** ∠ PEN= ∠ i / angle of incidence ∠ FEN'= ∠ r / angle of refraction ∠ SFM= ∠ e / angle of emergence ∠ FGH= ∠ D / angle of deviation (b) At face, AC of the prism $\frac{\sin i}{\sin r} = \frac{\sin \angle \text{EFM'}}{\sin \angle \text{SFM}} = \frac{2}{3}$ ∴ $\frac{\sin \angle \text{SFM}}{\sin \angle \text{EFM'}} = \frac{3}{2}$ Reason: Path of light is reversed at the face AC of the prism. ***Note:- Give one mark for part (b) to all students as in question given,*** $\frac{\sin \angle SFM}{\sin \angle EFM}$ ***is wrong. It should be*** $\frac{\sin \angle \text{SFM}}{\sin \angle \text{EFM'}}$	½ ½ ½ ½ 1	3
	SECTION D		
34.	(a) (i) A – Mouth/buccal cavity	½	

	B – Stomach C – Pancreas Enzymes A – Salivary Amylase B – Pepsin C – Trypsin/Lipase (ii) • Digested food is taken up by the walls of the small intestine/ villi • The food is absorbed here • Blood vessels in the villi take absorbed food to each and every cell of the body. • Utilisation of this food is done to obtain energy, building up new tissues and repair of old tissues. (if explained in paragraph award full marks) **OR** (b) (i) A – Pulmonary artery B – Aorta C – Capillaries **Functions:** A – Carries deoxygenated blood from heart to lungs. B – Carries oxygenated blood to different parts of the body. C- Exchange of material between blood and surrounding parts. (ii) • To prevent oxygenated and deoxygenated blood from mixing • This allows a highly efficient supply of oxygen to the body of birds to maintain their body temperature.	½ ½ ½ ½ ½ 2 ½ ½ ½ ½ ½ ½ 2	5
35.	(a) • X : 2,8,2 • Y: 2,8,7 • (Any one of the following below formations)  /	½ ½	

	 • Name: Magnesium chloride • Formula : $MgCl_2/XY_2$ • The compound will dissociate into ions / ions will move freely to the opposite electrodes/electrolysis will occur. • (i) Cathode: Mg^{2+} / Magnesium ion/Cation (ii) Anode: Cl^- /Chloride ion/Anion **OR** (b) (i) • Zinc, Iron, Lead, Copper (any two) • Zinc is more reactive than Iron/Iron is more reactive than Copper/Lead is more reactive than Copper (or any other suitable combination) • Demonstration: 1. Put iron nail in a solution of copper sulphate taken in a test tube. 2. After sometime blue colour of $CuSO_4$ fades. (Any other relevant activity) (ii) • Calcination • Reduction using Carbon Chemical Reactions 1. $ZnCO_3 \xrightarrow{Heat} ZnO + CO_2$ 2. $ZnO + C \longrightarrow Zn + CO$	1 ½ ½ 1 ½ ½ ½ , ½ 1 1 ½ ½ ½ ½	5
36	(a) (i) • Ohm's law: The potential difference, V, across the ends of a given metallic wire in an electric circuit is directly proportional to the current flowing through it, provided its temperature remains the same. / $V \alpha I$	1	

	• 	1	
	(ii)Here l = 1km = 1000 m R= 200Ω, A = 40 x 10^{-8} m^2	½	
	$\rho = \frac{R \times A}{l}$	½	
	$= 200\Omega \times \frac{40 \text{ x } 10^{-8} m^2}{1000 m}$	1	
	$= 8 \times 10^{-8}\ \Omega\, m$	1	
	OR		
	(b) (i) $R_s = R_1 + R_2 + R_3$	1	
	$\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$	1	
	(I) • Both have same reading. • As both are connected in series and in series current remains same.	½ ½	
	(II) $\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$ $= \frac{1}{30} + \frac{1}{30} + \frac{1}{30}$ =10 Ω	1	
	(III)		

	$I = \frac{V}{R} = \frac{3V}{30\Omega} = \frac{1}{10} = 0.1A$	1	5
	SECTION E		
37.	(a)  / CH_3CH_2OH	1	
	(b) Carbon dioxide and water / CO_2 and H_2O	½, ½	
	(c) (i)		
	• Ethanol is oxidised to ethanoic acid.	½	
	•  / CH_3COOH	1	
	• Acts as an oxidizing agent.	½	
	OR		
	(c) (ii)		
	• Ethene/ C_2H_4 is formed.	½	
	•  / $CH_2 = CH_2$	1	
	• Acts as a Dehydrating agent.	½	4
38.	(a) Sexual – Seed Asexual - Roots/stem/leaves. .	½ ½	
	(b) (i) Rose/Jasmine (or any other) (ii) Grapes/orange/banana (or any other)	½ ½	
	(c) (i) (Any two of the following) • Production of genetically similar plants • Propagated plants bear fruits and vegetables earlier than those produced from seeds. • Propagation of plants that have lost capacity to produce seeds. (Any other)	1, 1	
	OR		

	(c) (ii) Put the small pieces of potato into containing notches or buds in moist cotton for few days and observe. Notches or buds will give rise to fresh green shoots and roots.	2	4
39	(a) More than 2cm and less than 4cm from the optical centre of the lens (b) Less than 6cm from the optical centre of the lens (c) (i)  OB = 12 cm OF = 18 cm A′B′ : Image *[Fig. 9.16 (f) page 155 NCERT]* ***(Deduct ½ mark if direction of rays is not shown)*** **OR** (c) (ii) As per the information given in the question, the lens lies at the centre of the object placed and image formed.  The position of the object is at $2F_1$ therefore object distance = image distance = 2f ∴ 2f + 2f = 60 cm 4f = 60	1 1 2 1	

	f = 15cm ∴ The focal length of the lens = 15 cm **Note : Do not deduct marks if diagram is not drawn.**	1	4

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Marking Scheme
Strictly Confidential
(For Internal and Restricted use only)
Secondary School Supplementary Examination, July 2024
SUBJECT NAME: SCIENCE **SUBJECT CODE : 086**
PAPER CODE : 31/S/3

General Instructions: -

1	You are aware that evaluation is the most important process in the actual and correct assessment of the candidates. A small mistake in evaluation may lead to serious problems which may affect the future of the candidates, education system and teaching profession. To avoid mistakes, it is requested that before starting evaluation, you must read and understand the spot evaluation guidelines carefully.
2	**"Evaluation policy is a confidential policy as it is related to the confidentiality of the examinations conducted, Evaluation done and several other aspects. Its' leakage to public in any manner could lead to derailment of the examination system and affect the life and future of millions of candidates. Sharing this policy/document to anyone, publishing in any magazine and printing in News Paper/Website etc may invite action under various rules of the Board and IPC."**
3	Evaluation is to be done as per instructions provided in the Marking Scheme. It should not be done according to one's own interpretation or any other consideration. Marking Scheme should be strictly adhered to and religiously followed. **However, while evaluating, answers which are based on latest information or knowledge and/or are innovative, they may be assessed for their correctness otherwise and due marks be awarded to them. In class-X, while evaluating two competency-based questions, please try to understand given answer and even if reply is not from marking scheme but correct competency is enumerated by the candidate, due marks should be awarded.**
4	The Marking scheme carries only suggested value points for the answers.These are in the nature of Guidelines only and do not constitute the complete answer. The students can have their own expression and if the expression is correct, the due marks should be awarded accordingly.
5	The Head-Examiner must go through the first five answer books evaluated by each evaluator on the first day, to ensure that evaluation has been carried out as per the instructions given in the Marking Scheme. If there is any variation, the same should be zero after delibration and discussion. The remaining answer books meant for evaluation shall be given only after ensuring that there is no significant variation in the marking of individual evaluators.
6	Evaluators will mark(√) wherever answer is correct. For wrong answer CROSS 'X" be marked. Evaluators will not put right (✓)while evaluating which gives an impression that answer is correct and no marks are awarded. **This is most common mistake which evaluators are committing.**
7	If a question has parts, please award marks on the right-hand side for each part. Marks awarded for different parts of the question should then be totaled up and written in the left-hand margin and encircled. This may be followed strictly.
8	If a question does not have any parts, marks must be awarded in the left-hand margin and encircled. This may also be followed strictly.
9	If a student has attempted an extra question, answer of the question deserving more marks should be retained and the other answer scored out with a note **"Extra Question"**.
10	No marks to be deducted for the cumulative effect of an error. It should be penalized only once.

11	A full scale of marks 80 (example 0 to 80/70/60/50/40/30 marks as given in Question Paper) has to be used. Please do not hesitate to award full marks if the answer deserves it.
12	Every examiner has to necessarily do evaluation work for full working hours i.e., 8 hours every day and evaluate 20 answer books per day in main subjects and 25 answer books per day in other subjects (Details are given in Spot Guidelines).
13	Ensure that you do not make the following common types of errors committed by the Examiner in the past:- Giving more marks for an answer than assigned to it. ● Wrong totaling of marks awarded on an answer. ● Wrong transfer of marks from the inside pages of the answer book to the title page. Wrong question wise totaling on the title page. ● Leaving answer or part thereof unassessed in an answer book. ● ● Wrong totaling of marks of the two columns on the title page. ● Wrong grand total. ● Marks in words and figures not tallying/not same. ● Wrong transfer of marks from the answer book to online award list. ● Answers marked as correct, but marks not awarded. (Ensure that the right tick mark is correctly and clearly indicated. It should merely be a line. Same is with the X for incorrect answer.) ● Half or a part of answer marked correct and the rest as wrong, but no marks awarded.
14	While evaluating the answer books if the answer is found to be totally incorrect, it should be marked as cross (X) and awarded zero (0)Marks.
15	Any un assessed portion, non-carrying over of marks to the title page, or totaling error detected by the candidate shall damage the prestige of all the personnel engaged in the evaluation work as also of the Board. Hence, in order to uphold the prestige of all concerned, it is again reiterated that the instructions be followed meticulously and judiciously.
16	The Examiners should acquaint themselves with the guidelines given in the "**Guidelines for spot Evaluation**" before starting the actual evaluation.
17	Every Examiner shall also ensure that all the answers are evaluated, marks carried over to the title page, correctly totaled and written in figures and words.
18	The candidates are entitled to obtain photocopy of the Answer Book on request on payment of the prescribed processing fee. All Examiners/Additional Head Examiners/Head Examiners are once again reminded that they must ensure that evaluation is carried out strictly as per value points for each answer as given in the Marking Scheme.

SECONDARY SCHOOL SUPPLEMENTARY EXAMINATION, July 2024

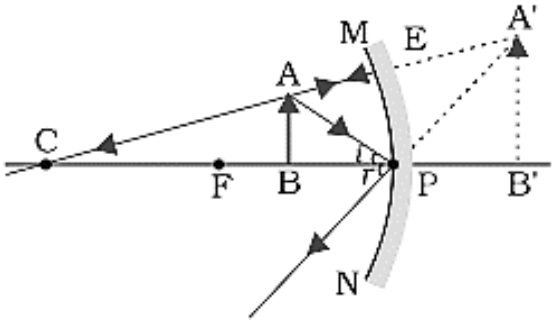
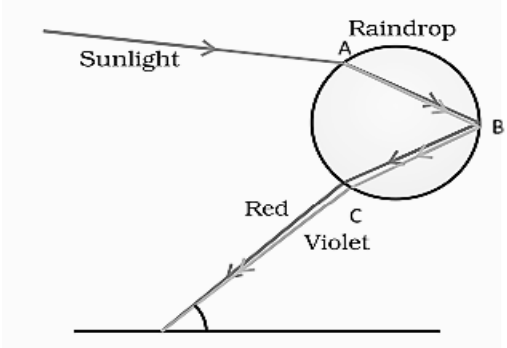
MARKING SCHEME

CLASS : X SCIENCE (Subject Code–086)

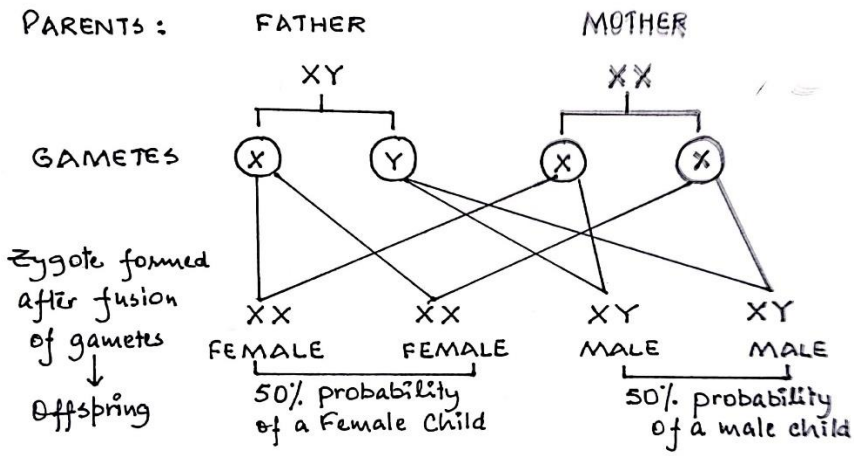
[Paper Code: 31/S/3]

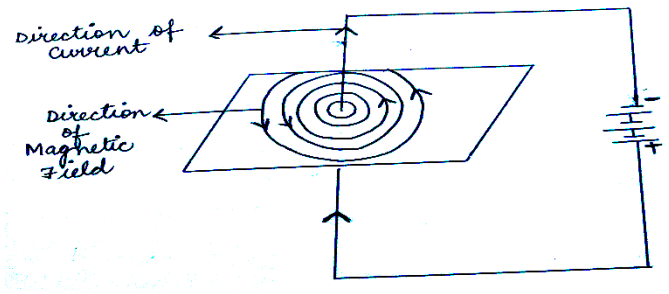
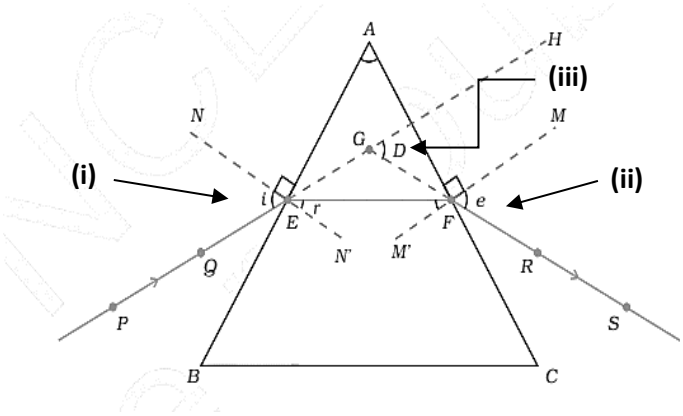
Maximum Marks: 80

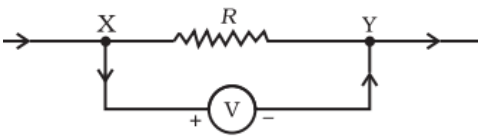
Q. No.	EXPECTED ANSWER / VALUE POINTS	Marks	Total Marks
	SECTION A		
1	(C)	**1**	**1**
2	(C)	**1**	**1**
3	(B)	**1**	**1**
4	(D)	**1**	**1**
5	(C)	**1**	**1**
6	(B)	**1**	**1**
7	(C)	**1**	**1**
8	(C)	**1**	**1**
9	(C)	**1**	**1**
10	(C)	**1**	**1**
11	(D)	**1**	**1**
12	(B)	**1**	**1**
13	(D)	**1**	**1**
14	(C)/(D) Both may happen.	**1**	**1**
15	(D)	**1**	**1**
16	(B)	**1**	**1**
17	(A)	**1**	**1**
18	(C)	**1**	**1**
19	(B)	**1**	**1**
20	(B)	**1**	**1**
	SECTION B		
21.	• Sodium • Sodium has more affinity for oxygen than carbon / Na is highly reactive • Electrolytic reduction	½ **1** ½	**2**

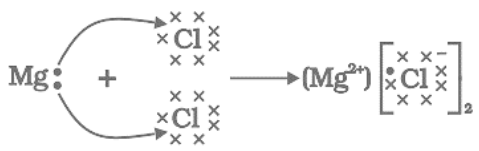
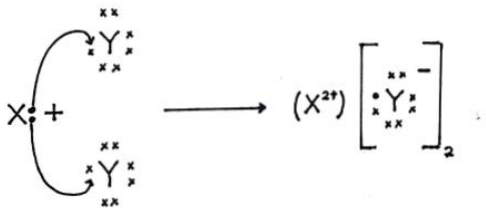
22.	 [P : Pole, F : Principal Focus, AB : Object, A′B′: Image] *[Diagram from NCERT fig 9.7(f) page 140]* Ray diagram Labelling of pole and principal focus Direction of rays	 1 ½ ½	 2
23.	(a)  *[Fig 10.8 page 16]* Diagram Correct labelling of A,B and C **OR** (b) • Ciliary muscles • Ciliary muscles adjust/change the focal length of eye lens by changing the curvature of eye lens.	 ½ 1½ 1 1	 2
24.	(a) • X – Sensory neuron • Y – Motor neuron • Functions: X : to carry message from skin/sense organs to brain / spinal cord Y : to carry response from brain/spinal cord to muscle / body parts **OR**	 ½ ½ ½ ½	

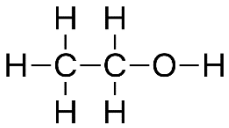
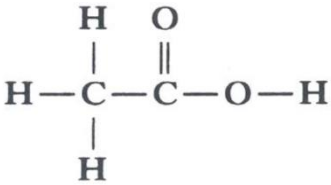
	(b) (i) Cerebellum/Hind brain (ii) Medulla/Hind brain (iii) Fore-brain **(iv)** Cerebrum/ Fore-brain	½ × 4	2
25.	• Two ways 1. Errors in DNA copying mechanism 2. Sexual reproduction • Importance: (any one of following) 1. Helps a species to survive 2. Helps an organism to adapt to the changing environment 3. Forms the basis of evolution **(or any other)**	½ ½ 1	2
26.	Because garden contain both biotic (grasses, trees, flowering plants, animals like frogs, insects and birds) components which interact with each other and with abiotic components (like temperature, rainfall, wind, soil and minerals) for their growth, reproduction and other activities.	**2**	**2**
	SECTION C		
27.	(a) $2Cu + O_2 \xrightarrow{\Delta} 2CuO$ Reddish brown Black (b) $CaO + H_2O \rightarrow Ca(OH)_2$ + heat/ rise in temperature (c) $BaCl_2 + Na_2SO_4 \rightarrow BaSO_4 + 2NaCl$ White precipitate ***(Or any other chemical reaction in each of the above cases)***	**1** **1** **1**	**3**
28.	**(**a) • Sodium carbonate/Na_2CO_3 • Recrystallisation of sodium carbonate / $Na_2CO_3 + 10H_2O \rightarrow Na_2CO_3.10H_2O$ • Basic. • Salt of a strong base and a weak acid. OR (b) • Sodium Hydroxide/NaOH/caustic soda – near cathode • Chlorine/Cl_2 – at anode • Hydrogen/H_2 – at cathode	**1** ½ ½ **1** ½ ½ ½	

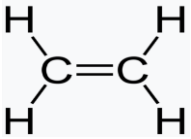
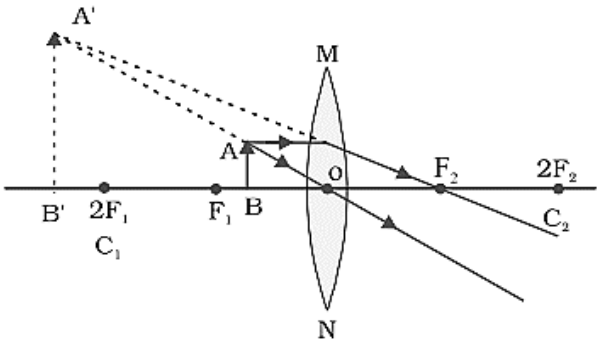
	Use: (any one use of each) • NaOH: Degreasing metal/soaps and detergents/ paper making/ artificial fibres. • Chlorine: water treatment/ PVC/ pesticides/ disinfectants/ CFCs • Hydrogen: fuel/ margarine/ammonia/fertilizers.	½ ½ ½	**3**
29.	(i) R = 2 Ω + 4 Ω + 6 Ω = 12 Ω (ii) $I = \frac{V}{R} = \frac{6V}{12\Omega} = 0.5$ A (iii) V= IR = 0.5A × 4Ω = 2V	1 1 1	3
30.	• Probability of male child born is 50% and female child born is 50% in human beings. •  **(Award full marks if explained in words)**	**1** **2**	**3**
31.	(a) (i) Sepals (ii) Petals (iii) Anther (iv) Ovule. (b) • Through the pollen tube.	½ ½ ½ ½ ½	

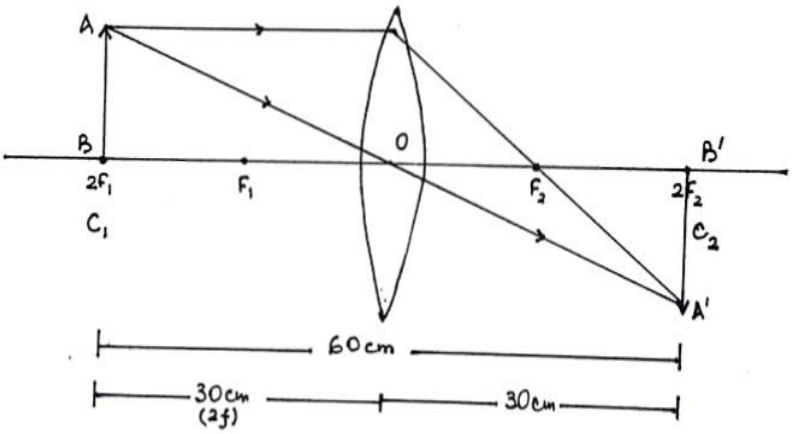
	• Male germ cell fuses with female germ cell to produce a zygote/ fertilization takes place	½	3
32.	**(a)**  Diagram Labelling of direction of current and corresponding field lines **(b)** (i) Fleming's Left Hand Rule (ii) Out of the plane of paper	1 1 ½ ½	**3**
33.	•  Labelled diagram *[Fig 10.4, page 166 of NCERT]* • White light will split into a band of seven colours of spectrum. • Different colours of white light bend / refract through different angles with respect to incident ray as they pass through a prism.	1 ½ ½ 1	3
	SECTION D		
34.	(i) • R= $\rho \frac{l}{A}$		

	$\rho = \frac{R \times A}{l}$	1	
	• SI unit of $\rho = \frac{\text{SI unit of R} \times \text{SI unit of A}}{\text{SI unit of l}} = \frac{\Omega \times m^2}{m}$		
	$= \Omega$ m	1	
	(ii)R= 3Ω ; l =120 cm = 1.2 m ;		
	d = 0.4 mm or r = 0.2 mm	½	
	$\rho = \frac{R \times A}{l}$		
	A= $3.14 \times (0.2mm)^2 = 3.14 \times 4 \times 10^{-8} m^2$	½	
	= 3Ω x $\frac{3.14 \times 4 \times 10^{-8} \ m^2}{1.2m}$	1	
	= 3.14×10^{-7} Ω m	1	
	OR		
	(i)		
	• When electric current is passed through a resistor electric energy is dissipated and appears as heat energy.	1	
	• When charge Q flows through a conductor of resistance R having potential difference of V , the workdone W is given by-		
			
	W= V x Q		
	= V x I x t		
	= IR x I x t = I^2Rt	1	
	(∵V = IxR – Ohm's Law)		
	When circuit is purely resistive electric energy is converted into heat energy.		
	H= I^2Rt	1	
	(ii) Here I= 2A ; R= 10Ω ; t= 100s		
	H= I^2Rt		
	= $(2A)^2$ x 10Ω x 100s	1	

	= 4000 J	1	5
35.	(a). (i) A – Pyruvate B - Ethanol C – Muscle cells D – Mitochondria (ii) (I) Plants – Starch (II) Animals – Glycogen (iii) Due to lack of oxygen, pyruvic acid is converted to lactic acid which accumulates in their muscle cells. so causes cramps. (iv) Amount of dissolved oxygen in water is low as compared to amount of oxygen in air, so the rate of breathing is more to get required oxygen in aquatic animals. **OR** (b) (i) A – Chloroplast B – Guard cells Functions: A – contain chlorophyll pigment to absorb light during photosynthesis. B – Regulate opening and closing of stomatal pore. (ii) Xylem and Phloem Functions: Xylem – transport water and minerals from soil to other body parts Phloem – Transport carbohydrates as food from leaves to other body parts / translocation (iii) Uncovered part of leaf receive light during photosynthesis and produce starch which on reacting with iodine make the leaf blue black in colour.	½ ½ ½ ½ ½ ½ **1** **1** ½ ½ ½ ½ ½ +½ ½ ½ **1**	**5**
36.	(a) • X : 2,8,2 • Y: 2,8,7 • (Any one of the following below formations)  /  • Name: Magnesium chloride • Formula : $MgCl_2$/XY_2	½ ½ 1 ½ ½	

	• The compound will dissociate into ions/ions will move freely to the opposite electrodes/electrolysis will occur.	1	
	• (i) Cathode: Mg^{2+} / Magnesium ion/Cation (ii) Anode: Cl^- /Chloride ion/Anion	½ ½	
	OR		
	(b) (i) • Zinc, Iron, Lead, Copper (any two) • Zinc is more reactive than Iron/Iron is more reactive than Copper/Lead is more reactive than Copper (or any other suitable combination) • Demonstration: 1. Put iron nail in a solution of copper sulphate taken in a test tube. 2. After sometime blue colour of $CuSO_4$ fades. (Any other relevant activity)	½ , ½ 1 1	
	(ii) • Calcination • Reduction using Carbon Chemical Reactions 1. $ZnCO_3 \xrightarrow{Heat} ZnO + CO_2$ 2. $ZnO + C \longrightarrow Zn + CO$	 ½ ½ ½ ½	5
	SECTION E		
37.	(a)  / CH_3CH_2OH	1	
	(b) Carbon dioxide and water / CO_2 and H_2O	½, ½	
	(c) (i) • Ethanol is oxidised to ethanoic acid.	 ½	
	•  / CH_3COOH	1	
	• Acts as an oxidizing agent.	½	
	OR		

	(c) (ii) • Ethene/ C_2H_4 is formed. •  / $CH_2 = CH_2$ • Acts as a Dehydrating agent.	½ 1 ½	4
38.	(a) Sexual – Seed Asexual - Roots/stem/leaves. . (b) (i) Rose/Jasmine (or any other) (ii) Grapes/orange/banana (or any other) (c) (i) (Any two of the following) • Production of genetically similar plants • Propagated plants bear fruits and vegetables earlier than those produced from seeds. • Propagation of plants that have lost capacity to produce seeds. (Any other) **OR** (c) (ii) Put the small pieces of potato into containing notches or buds in moist cotton for few days and observe. Notches or buds will give rise to fresh green shoots and roots.	½ ½ ½ ½ 1, 1 2	4
39	(a) More than 2cm and less than 4cm from the optical centre of the lens (b) Less than 6cm from the optical centre of the lens (c) (i)  	1 1 2	

	OB = 12 cm OF = 18 cm A′B′ : Image *[Fig. 9.16 (f) page 155 NCERT]* ***(Deduct ½ mark if direction of rays is not shown)*** **OR** (c) (ii) As per the information given in the question, the lens lies at the centre of the object placed and image formed.  The position of the object is at $2F_1$ therefore object distance = image distance = 2f ∴ 2f + 2f = 60 cm 4f = 60 f = 15cm ∴ The focal length of the lens = 15 cm **Note : Do not deduct marks if diagram is not drawn.**	1 1	4

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Marking Scheme
Strictly Confidential
(For Internal and Restricted use only)
Secondary School Supplementary Examination, July 2024
SUBJECT NAME: SCIENCE **SUBJECT CODE: 086**
PAPER CODE: 31/B

General Instructions: -

1	You are aware that evaluation is the most important process in the actual and correct assessment of the candidates. A small mistake in evaluation may lead to serious problems which may affect the future of the candidates, education system and teaching profession. To avoid mistakes, it is requested that before starting evaluation, you must read and understand the spot evaluation guidelines carefully.
2	**"Evaluation policy is a confidential policy as it is related to the confidentiality of the examinations conducted, Evaluation done and several other aspects. Its' leakage to public in any manner could lead to derailment of the examination system and affect the life and future of millions of candidates. Sharing this policy/document to anyone, publishing in any magazine and printing in News Paper/Website etc may invite action under various rules of the Board and IPC."**
3	Evaluation is to be done as per instructions provided in the Marking Scheme. It should not be done according to one's own interpretation or any other consideration. Marking Scheme should be strictly adhered to and religiously followed. **However, while evaluating, answers which are based on latest information or knowledge and/or are innovative, they may be assessed for their correctness otherwise and due marks be awarded to them. In class-X, while evaluating two competency-based questions, please try to understand given answer and even if reply is not from marking scheme but correct competency is enumerated by the candidate, due marks should be awarded.**
4	The Marking scheme carries only suggested value points for the answers.These are in the nature of Guidelines only and do not constitute the complete answer. The students can have their own expression and if the expression is correct, the due marks should be awarded accordingly.
5	The Head-Examiner must go through the first five answer books evaluated by each evaluator on the first day, to ensure that evaluation has been carried out as per the instructions given in the Marking Scheme. If there is any variation, the same should be zero after delibration and discussion. The remaining answer books meant for evaluation shall be given only after ensuring that there is no significant variation in the marking of individual evaluators.
6	Evaluators will mark(√) wherever answer is correct. For wrong answer CROSS 'X" be marked. Evaluators will not put right (✓)while evaluating which gives an impression that answer is correct and no marks are awarded. **This is most common mistake which evaluators are committing.**
7	If a question has parts, please award marks on the right-hand side for each part. Marks awarded for different parts of the question should then be totaled up and written in the left-hand margin and encircled. This may be followed strictly.
8	If a question does not have any parts, marks must be awarded in the left-hand margin and encircled. This may also be followed strictly.
9	If a student has attempted an extra question, answer of the question deserving more marks should be retained and the other answer scored out with a note **"Extra Question"**.
10	No marks to be deducted for the cumulative effect of an error. It should be penalized only once.

11	A full scale of marks 80 (example 0 to 80/70/60/50/40/30 marks as given in Question Paper) has to be used. Please do not hesitate to award full marks if the answer deserves it.
12	Every examiner has to necessarily do evaluation work for full working hours i.e., 8 hours every day and evaluate 20 answer books per day in main subjects and 25 answer books per day in other subjects (Details are given in Spot Guidelines).
13	Ensure that you do not make the following common types of errors committed by the Examiner in the past:- Giving more marks for an answer than assigned to it. ● Wrong totaling of marks awarded on an answer. ● Wrong transfer of marks from the inside pages of the answer book to the title page. Wrong question wise totaling on the title page. ● Leaving answer or part thereof unassessed in an answer book. ● ● Wrong totaling of marks of the two columns on the title page. ● Wrong grand total. ● Marks in words and figures not tallying/not same. ● Wrong transfer of marks from the answer book to online award list. ● Answers marked as correct, but marks not awarded. (Ensure that the right tick mark is correctly and clearly indicated. It should merely be a line. Same is with the X for incorrect answer.) ● Half or a part of answer marked correct and the rest as wrong, but no marks awarded.
14	While evaluating the answer books if the answer is found to be totally incorrect, it should be marked as cross (X) and awarded zero (0)Marks.
15	Any un assessed portion, non-carrying over of marks to the title page, or totaling error detected by the candidate shall damage the prestige of all the personnel engaged in the evaluation work as also of the Board. Hence, in order to uphold the prestige of all concerned, it is again reiterated that the instructions be followed meticulously and judiciously.
16	The Examiners should acquaint themselves with the guidelines given in the "**Guidelines for spot Evaluation**" before starting the actual evaluation.
17	Every Examiner shall also ensure that all the answers are evaluated, marks carried over to the title page, correctly totaled and written in figures and words.
18	The candidates are entitled to obtain photocopy of the Answer Book on request on payment of the prescribed processing fee. All Examiners/Additional Head Examiners/Head Examiners are once again reminded that they must ensure that evaluation is carried out strictly as per value points for each answer as given in the Marking Scheme.

SECONDARY SCHOOL SUPPLEMENTARY EXAMINATION, July 2024

MARKING SCHEME

CLASS : X SCIENCE (Subject Code–086)

[Paper Code: 31/B]

Maximum Marks: 80

Q. No.	EXPECTED ANSWERS / VALUE POINTS	Marks	Total Marks
1	(B)	1	1
2	(A)	1	1
3	(C)	1	1
4	(D)	1	1
5	(C)	1	1
6	(D)	1	1
7	(D)	1	1
8	(D)	1	1
9	(A)	1	1
10	(D)	1	1
11	(C)	1	1
12	(D)	1	1
13	(B)	1	1
14	(A)	1	1
15	(B)	1	1
16	(C)	1	1
17	(A)	1	1
18	(C)	1	1
19	(D)	1	1
20	(A) / (B) R-statement is right the explanation for only villi, but not for blood vessels.	1	1
	SECTION B		
21	Ozone shields the surface of the earth to protect from the harmful effects of the ultra violet radiations emitted from the sun. These radiations can cause skin cancer in human beings, damages to organisms.	2	2
22	(a) • Calcium sulphate hemihydrate • $CaSO_4.\frac{1}{2}H_2O$	 ½ ½	

	• To prevent its reaction with moisture / from forming a hard mass (gypsum) **OR** (b) • The reaction between an acid and a base to form salt and water. • $NaOH + HCl \rightarrow NaCl + H_2O$ (Sodium chloride) / (salt) **(Or any other suitable reaction)**	1 1 1	2
23	(i) • Bile juice in small intestine converts acidic food of stomach into alkaline for the pancreatic enzyme to act • Bile salts allow breakdown/emulsification of large fat globules into smaller droplets. (ii)Role of pancreatic enzymes: • Digestion of proteins into amino acids. • Breaking down of emulsified fats into fatty acids and glycerol. (Any other suitable role)	½ ½ ½ ½	2
24	• Plant hormones: These are the chemical compounds in plants which help to coordinate growth, development, and responses to the environment. • (i) Auxin (ii) Cytokinin	1 ½ ½	2
25	(a) (i) Myopia (ii) • Excessive curvature of eye lens / Converging power of eye lens increases • Elongation of the eye ball. (ii) Diverging lens	½ ½ ½ ½	

	OR (b) • Very fine particles of the atmosphere scatter short wavelength colour, mainly blue light, but scattering is not prominent at very high altitudes. So, sky appears black.	1+1	2
26	(a) • Vegetative propagation: An asexual method of growing new plants from the vegetative parts of the plants such as root, stem or leaf. • Advantages: (any two) • Plants produced are genetically similar to the parent plant. • Plants produced bear flowers and fruits earlier than those produced from seeds. • Propagation of seedless plants. **OR** (b) (any two)	1 ½ + ½	

Sexual reproduction	Asexual reproduction
Two parents (male and female) involved	Only one parent involved
Gametes are involved	Gametes are not involved
Zygote is formed	Zygote is not formed
More variations are seen in the offsprings.	Less variations are seen in the offsprings.

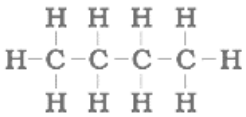
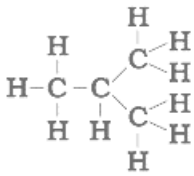
1+1 | 2

	SECTION C		
27	(a) • pH = 7 • Reason: It is a neutral salt formed by NaOH (strong base) and HCl (strong acid). (b) • Orange • Reason: salt of a weak base and strong acid. (c) • No change/Remains blue	½ ½ ½ ½ ½	

	• Reason: basic salt formed by strong base and a weak acid	½	3
28	(a) (i) Cathode – Hydrogen (ii) Anode – Oxygen (b) As number of molecules of hydrogen gas are twice of number of molecules of oxygen gas, so the volume of H_2 is two times of O_2(gas). (c) (Test for anyone gas) • Test for hydrogen gas: Bring a lighted splinter at the mouth of the test tube containing hydrogen gas, gas burns with pop sound/hiss. • Test for oxygen gas: Insert a glowing splinter into the test tube containing oxygen gas, this splinter reignites.	½ ½ 1 1	3
29	(a) (i) Two characteristics: • Presence of alveoli • Extensive network of blood vessels (or any other) (ii) (1) Nasal passage (2) Diaphragm (3) Alveoli (4) Bronchi (Bronchus) **OR** (b) (i) (1) Kidney (2) Urinary bladder (3) Ureter (4) Urethra Two factors: • Amount of excess water in the body • Amount of dissolved wastes to be excreted	½ + ½ 4 x ½ 4 x ½ ½ ½	3
30	• Plants (green) → Goat → Human Being (or any other food chain) • Pesticides are washed down into the soil and water bodies. From the soil pesticides are absorbed by crop plants and enter the food chain. These chemicals are non-biodegradable and get accumulated progressively at each trophic level.	1 1	

	• As human beings occupy the top level in any food chain, the maximum concentration of these chemicals gets accumulated in our bodies.	1	3
31	(a) (i) Here h = 5 cm; f = + 20 cm; u = - 30 cm Lens formula $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$ $\frac{1}{v} - \frac{1}{-30} = \frac{1}{20}$ $\frac{1}{v} + \frac{1}{30} = \frac{1}{20}$ $v =$ + 60 cm, image distance is 60 cm from the lens. (ii) $h' = \frac{v}{u} \times h$ $= \frac{+60\text{ cm}}{-30\text{ cm}} \times 5\text{ cm}$ = -10 cm, height of the image is 10 cm. **OR** (b) • Converging or diverging ability of a lens / Reciprocal of focal length of a lens in metre / P $= \frac{1}{f(m)}$ • ➢ Converging ➢ P = $\frac{100}{+25}$ = +4D • Same size • Real • Inverted (any two)	½ ½ 1 ½ ½ 1 ½ ½ ½ + ½	3

32	• Splitting of white light into its component colours when it passes through a prism/rain drop/oil bubble. • Red colour bends the least • Violet colour bends the most. • Different colours of white light bend/refract through different angles with respect to the incident ray as they pass through a prism. **OR** (b) • ➢ After the rain shower. ➢ In a direction opposite to that of the sun. • The tiny water droplets ➢ refract and disperse sunlight then ➢ reflect it internally and finally ➢ refract it again when it comes out of the rain drop. • Two necessary condition – (i) Presence of tiny rain droplets in air. (ii) Sun behind the observer.	1 ½ ½ 1 ½ ½ 1 ½ + ½	3
33	(a) Current used in household supply is AC (alternating current) which changes its direction periodically with time whereas the current given by a battery of dry cells is DC (direct current) which does not change its direction with time. (b) • Alternating current • Alternating current can be transmitted over long distances without much loss of energy. (c) At the time of short circuiting, current in the circuit exceeds the rated value of the fuse, hence the fuse melts and breaks the circuit.	1 ½ ½ 1	3
	SECTION D		
34	(a) (i) (1) Ethanol; C_2H_5OH (2) Ethanoic Acid; CH_3COOH	 ½ + ½ ½ + ½	

	(ii) (1) • An ester is formed • $C_2H_5OH + CH_3COOH \xrightarrow[warm]{Acid} CH_3COOC_2H_5 + H_2O$ Ester (2) • Ethene is formed / $C_2H_5OH \xrightarrow[Excess\ Conc\ H_2SO_4]{443\ K} C_2H_4 + H_2O$ • Conc. H_2SO_4 acts as a dehydrating agent. **OR** (b) (i) • Carbon compounds having same molecular formula but different structures are called structural isomers  / $CH_3—CH_2-CH_2-CH_3$  / $CH_3—CH(CH_3)—CH_3$ (ii) • Methane, CH_4 • C_nH_{2n+2} • C_2H_6 • $CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$ + heat	½ 1 1 ½ 1 ½ ½ ½ ½+½ ½ ½ 1	5
35	(a)		

	(i) Function of: (1) Stigma: To trap pollen grains. (2) Pollen tube: To transfer male gamete to female gamete. (3) Anther: To produce pollen grains. (ii) (1) Plumule (2) Radicle (iii) (1) Ovule (2) Ovary **OR** (b) (i) • Testes • Testosterone • Function of testosterone: To bring about changes in appearance seen in boys at the time of puberty/Regulate the formation of sperms. (or any other) (ii) (i) Oviduct / Fallopian Tube (ii) Uterus (iii) • The embryo gets nutrition from the mother's blood with the help of a special tissue called placenta which connects growing embryo with the uterine wall. • It acts as a site for exchange of gases and transfer of nutrients from mother's blood to embryo and to remove wastes from embryo to mother's blood.	1 1 1 ½ ½ ½ ½ ½ ½ 1 ½ ½ 2	5
36	(a) • Two factors (any two) (i) Length (l) (ii) area of cross-section (A) (iii) Material of conductor • $R = \rho \frac{l}{A}$ or $\rho = R\, l/A$ • SI unit of ρ = SI unit of (R) × $\frac{\text{SI unit of (A)}}{\text{SI unit of (l)}}$ $= \text{ohm} \times \frac{\text{metre}^2}{\text{metre}} = \text{ohm . metre}$ (b) $\rho = R\, l/A$ $= 40\ \Omega \times \frac{6.5 \times 10^{-8}\ \text{m}^2}{1\ \text{m}}$ $= 26 \times 10^{-7}\ \Omega\text{m}$	½ + ½ 1 1 1 1	5

	SECTION E		
37	(a) A homogeneous mixture of two or more metals or metals and nonmetals. (b) Copper and tin. (c) (i) A homogeneous mixture of Iron, Chromium and Nickel, formed by melting and dissolving these three metals in a proper ratio. • ➢ Stainless steel is harder than iron. ➢ It does not rust. **OR** (c) (ii) • Solder is an alloy of lead and tin. • It is used in welding of electric wires. ➢ Because it has low melting point	1 1 1 ½ ½ 1 ½ ½	4
38	(a) **(Any two)** (i) Round / wrinkled seeds (ii) White / violet flowers (iii) Yellow and green seeds (b) Dominant trait: A trait or gene which expresses itself even in the presence of its contrasting trait Recessive trait: A trait (gene) which cannot express itself in the presence of its dominant trait. (c) (i) • Tall plants: Dwarf plants 3 : 1 • As both the copies of recessive trait reappear in a the F_2 generation **OR** (c) (ii) (any two of the following) **F_1 generation**: Progeny of parental plants/first generation plants is called F_1 generation **F_2 generation**: Progeny of first-generation plants /second generation plants is called F_2 generation Obtained by crossing/cross pollinating parental plants Obtained by selfing /self pollination of F1 generation Only dominant trait/tall plants appear Both dominant and recessive traits may appear/both tall and dwarf pea plants are present.	½ + ½ 1 1 1 1+1	4

39	(a) It is a point on the principal axis of the mirror where rays incident, parallel to principal axis will converge after reflection. (b) R =2f = 2 x 12 cm = 24 cm (c) (i) (1) Image is • real • inverted and • magnified (Any two) (3) Image is • virtual • erect and • magnified (Any two) **OR** **(c)** **(ii)** (1) In this case, the object is at C, therefore R = 40 cm R=2f ∴ Focal length (f) = $\frac{40}{2}\,cm$ = 20 cm (2) Magnification(m) = -1 (Image of same size and inverted)	1 1 ½ + ½ ½ + ½ ½ ½ 1	4